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Numerical and experimental study on spatter formation mechanisms and suppression method during ring-mode laser spiral scanning welding of T2 copper

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Article

Abstract

Ring-mode laser spiral scanning welding (RMLSSW) is an innovative technology for low-spatter manufacturing of thin-walled components in new energy systems. However, the spatter-free process window of RMLSSW remains empirical. This study integrates experiments and simulations to elucidate how spatiotemporal energy distribution governs spatter initiation and suppression. The spatter formation mechanism of RMLSSW is initiated by insufficient preheating, which causes sharp absorptivity fluctuations. These fluctuations generate steep thermal gradients and transverse molten pool vortices that shear the keyhole, culminating in violent evaporation at the keyhole crown and

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subsequent spatter ejection. Power of ring (P_r) and spiral pitch (d) are identified as the critical parameters for achieving spatter suppression. Simulations reveal that increasing P_r or decreasing d raises substrate temperature and laser absorptivity, enlarges the molten pool, stabilizes melt flow around the keyhole, and reduces the probability of spatter initiation. The enhanced recoil pressure dilates the keyhole, increasing its depth-to-diameter ratio and volume, thereby suppressing spatter induced by stochastic keyhole collapse. An optimal preheating process window of $P_r=2000\text{--}2400\text{ W}$ and $d=0.55\text{--}0.70\text{ mm}$ is proposed, within which laser–material coupling is maximized and spatter is minimized. This study provides transferable theoretical foundations and well-defined optimization criteria for achieving spatter-free, high-quality laser welding of highly reflective thin-walled materials.

Keywords

Ring-mode laser spiral scanning welding; T2 copper; Spatter formation mechanism; Spatter suppression mechanism

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1. Introduction

Owing to superior electrical and thermal conductivities, copper is widely used in critical components of new energy vehicles [1]. Laser scanning welding, offering high energy efficiency, low heat input, and exceptional process flexibility, has become the predominant joining technology for thin-walled copper structures [2]. However, copper's low infrared absorptivity at room temperature and high thermal conductivity cause rapid coupling efficiency fluctuations during welding [3], destabilizing the keyhole and triggering violent spatter [4], [5]. Such spatter degrades weld appearance and risks battery contamination, potentially causing short circuits and safety incidents [6].

[Declaration of competing interest](#)

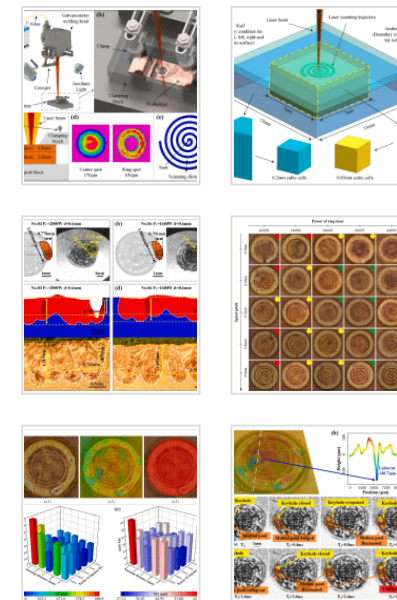
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Realizing low-spatter, high-reliability welding of copper has therefore become a key challenge in new energy vehicle manufacturing.

Table 4. Experimental validation param...

Recent spatter suppression efforts for highly reflective materials include blue laser [7], [8], [9], surface modification [10], [11], scanning laser welding [12], [13], [14], and adjustable ring-mode laser welding [15]. Scanning and ring-mode processes attract particular attention due to their industrial maturity and efficacy. In scanning-laser welding, Mei et al. [16] demonstrated that beam oscillation prolongs the keyhole life cycle and enhances process stability. Wan et al. [17] found that spatter probability drops sharply when oscillation frequency exceeds the keyhole's natural frequency. Wang et al. [18] further revealed that low-frequency oscillation suppresses spatter, while high frequency promotes micro-droplet ejection. In adjustable ring-mode welding, Li et al. [19] found that the high-energy core ensures penetration while the ring stabilizes the molten pool. Zhang et al. [20] showed that the outer ring beam reduces the temperature gradient and weakens the surface-tension gradient, yielding smoother molten pool flow, lower liquid-column height and frequency, and less spatter. Lassila et al. [21] demonstrated that an auxiliary ring beam increases keyhole recoil pressure and enlarges its aperture, while Wang et al. [22] revealed that ring laser irradiation suppresses periodic keyhole collapse, markedly reducing spatter. Miana et al. [23] established that core power enables deep penetration, while ring power ensures favorable temperature distribution for both quality and depth. Wang et al. [24] observed that appropriate ring power widens and stabilizes the keyhole, improving weld surface quality and reducing spatter. Mohammadpour et al. [25] found that a higher ring-beam power fraction increases penetration depth while minimizing spatter formation. Li et al. [26] confirmed that ring-beam welding effectively suppresses spatter and offers a broad process window. Complementarily, Yang et al. [27] elucidated a synergistic mechanism wherein the ring-mode enlarges the keyhole opening while oscillation induces stable vortices, fundamentally suppressing collapse-induced spatter. Collectively, these studies show that spatiotemporal coupling of ring-mode and scanning trajectories regulates keyhole dynamics and stabilizes melt flow, offering an effective spatter suppression route for highly reflective materials. However, systematic

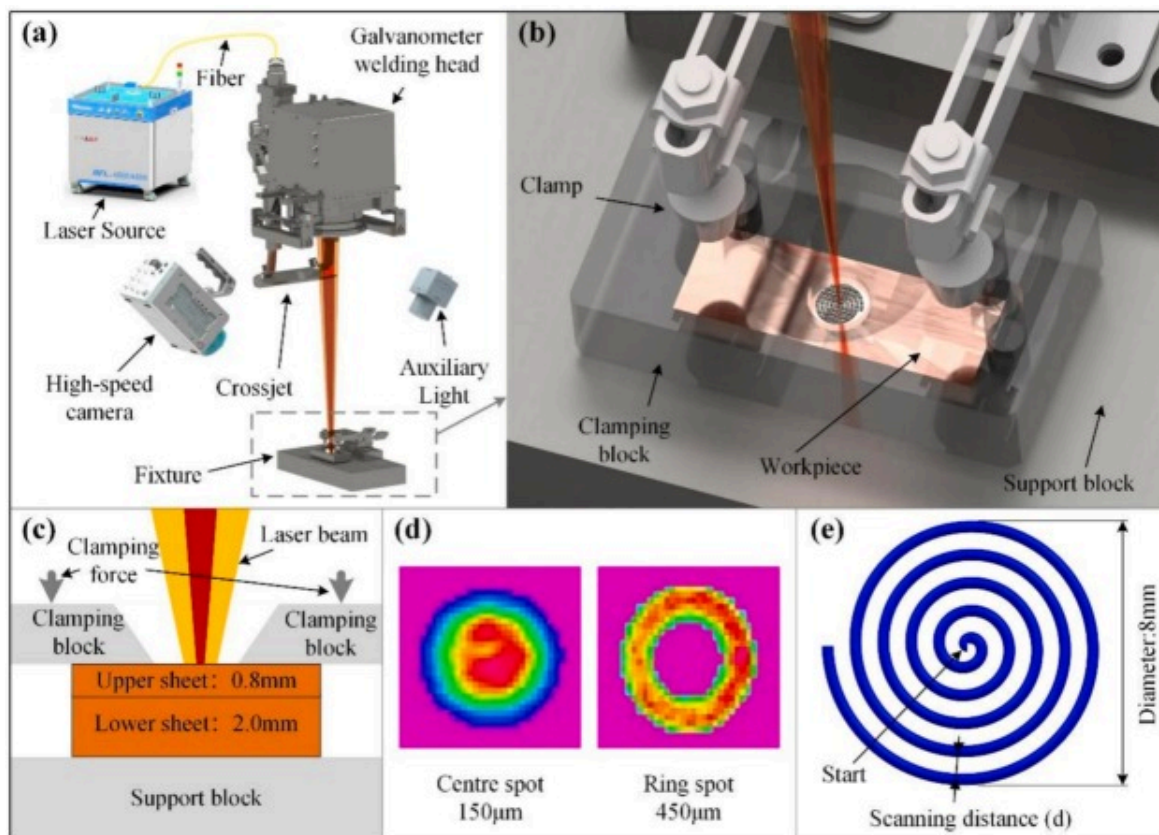
investigations of ring-mode scanning welding for copper remain limited. The quantitative laws governing laser energy distribution throughout spatter initiation, evolution, suppression require further elucidation.

In this work, RMLSSW of T2 copper is studied experimentally and numerically. A 3D transient CFD model clarifies how the power of ring (P_r) and spiral pitch (d) suppress keyhole instability and melt ejection. An optimized spatter-suppression strategy based on spatiotemporal energy distribution is proposed and validated, providing theoretical and process guidelines for spatter-free welding of thin-walled highly reflective materials.

2. Materials and methods

2.1. Experimental setup

Fig. 1(a) schematically illustrates the experimental setup. An adjustable beam profile laser (Raycus ABP RFL4000/4000) served as the power source, delivering a maximum of 4 kW in the central beam and 4 kW in the ring beam, with a combined output of 8 kW. The emission wavelength was 1080 ± 5 nm, the central and ring fibers possessed core diameters of 50 μm and 150 μm , respectively. The beam was projected through a galvanometric scanning head (collimation length 125 mm, focal length 420 mm) capable of a maximum scan speed of 9000 mm/s and of generating a focused spot with a diameter of 150 μm at the center and 450 μm in the ring (**Fig. 1(d)**). A high-speed camera (Phantom VEO 610L) oriented at an angle of 60° to the surface normal recorded keyhole and molten pool dynamics at a frame rate of 5,000 fps with a spatial resolution of 960×960 pixels. A diode laser illumination system (Cavitar, 808 nm wavelength) was employed as an auxiliary illumination source to minimize image degradation caused by plasma luminescence and metal vapor glare, enabling clear visualization of the molten pool and keyhole dynamics.



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Fig. 1. Experiment setup. (a) overall experimental setup, (b) experimental fixture, (c) schematic of plate thickness and clamping, (d) focal spot size, (e) laser scanning trajectory.

The substrate material was T2 copper (composition provided in Table 1). Lap joints were produced between an upper sheet (60 mm × 20 mm × 0.8 mm) and a lower sheet (60 mm × 20 mm × 2 mm). Prior to welding, the faying surfaces were degreased with anhydrous ethanol and the assembly was rigidly clamped using an aluminum block and

quick-action clamp (Fig. 1(b) and(c)). The laser beam was delivered normal to the surface without a shielding gas. A cross-jet air curtain protected the optics from ejected spatter.

Table 1. Chemical composition of T2 copper.

Chemical composition	Cu	Bi	Sb	As	Fe	Pb	S
Mass fraction/%	99.90	0.001	0.002	0.002	0.005	0.005	0.005

Previous studies have established that in the adjustable ring laser welding, the central beam is instrumental in achieving adequate deep penetration, whereas the ring beam markedly enhances the stability of the molten pool and keyhole [21]. Laser power and spiral pitch exert a significant coupled effect on weld formation quality and porosity suppression [28], [29]. Building on these findings, this study employed a constant power of center (P_c) of 3000 W and a welding velocity (v) of 200 mm/s. The welding trajectory was an Archimedean spiral, welding from the inside out with an outer diameter of 8 mm, as depicted in Fig. 1(e). This study specifically examines the influence of the power of ring (P_r) and spiral pitch (d) on welding spatter behavior. The detailed experimental parameters are presented in Table 2. Critical metrics, including spatter-induced crater depth and surface roughness, were obtained by integrating height cloud maps and cross-sectional profile data, followed by meticulous measurement and statistical analysis.

Table 2. Experimental parameters.

No.	Power of center (P_c , W)	Power of ring (P_r , W)	Spiral pitch (d , mm)	Welding velocity (v , mm/s)
1	3000	1600	0.5	200
2	3000	1800	0.5	200
3	3000	2000	0.5	200

No.	Power of center (P_c , W)	Power of ring (P_r , W)	Spiral pitch (d , mm)	Welding velocity (v , mm/s)
4	3000	2200	0.5	200
5	3000	2400	0.5	200
6	3000	1600	0.6	200
7	3000	1800	0.6	200
8	3000	2000	0.6	200
9	3000	2200	0.6	200
10	3000	2400	0.6	200
11	3000	1600	0.7	200
12	3000	1800	0.7	200
13	3000	2000	0.7	200
14	3000	2200	0.7	200
15	3000	2400	0.7	200
16	3000	1600	0.8	200
17	3000	1800	0.8	200
18	3000	2000	0.8	200
19	3000	2200	0.8	200
20	3000	2400	0.8	200
21	3000	1600	0.9	200
22	3000	1800	0.9	200
23	3000	2000	0.9	200

No.	Power of center (P_c , W)	Power of ring (P_r , W)	Spiral pitch (d , mm)	Welding velocity (v , mm/s)
24	3000	2200	0.9	200
25	3000	2400	0.9	200

2.2. Numerical simulation

2.2.1. Numerical implementations

A fully coupled three-dimensional CFD model was developed to simulate the adjustable ring laser welding [15], [30], [31], [32]. The model solves the conservation equations of mass, momentum, and energy to obtain the transient velocity and temperature fields of the molten pool. The evolving keyhole geometry is captured using volume-of-fluid (VOF) scheme [33], [34], and multiple laser reflections and absorptions on the keyhole wall are handled by a ray-tracing algorithm. This study adapts and extends this modeling framework to RMLSSW of T2 copper to elucidate the underlying mechanisms of spatter formation.

The trajectory of the laser spot is prescribed at a constant linear velocity v with a spiral pitch d . At $t=0$ the spot is located at the coordinate origin and moves in the positive x-direction, following the parametric equations of an Archimedean spiral,

$$r(\theta) = \frac{d}{2\pi} \bullet \theta \quad (1)$$

The instantaneous velocity direction angle is $\theta(t)$,

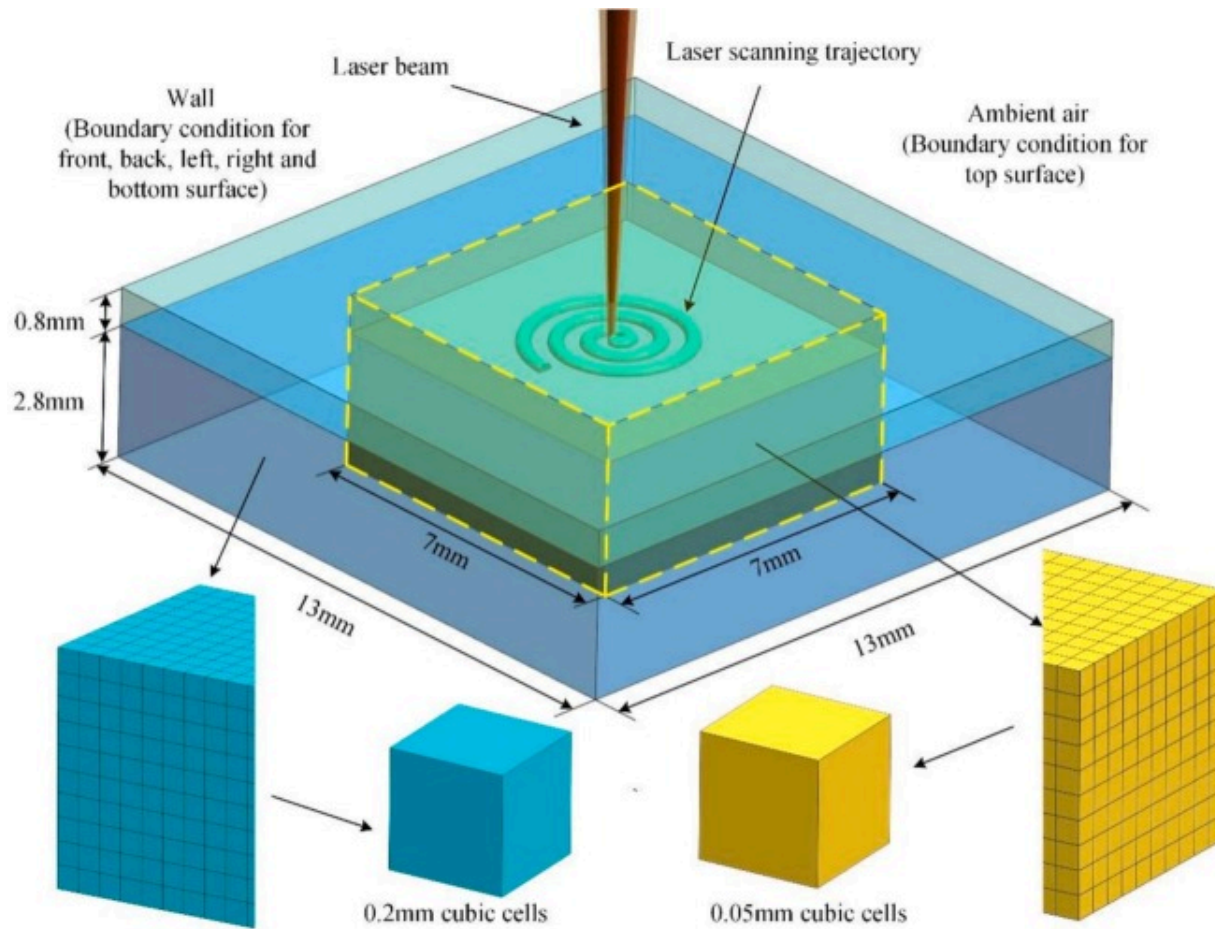
$$\theta(t) = \sqrt{\frac{4\pi v \bullet t}{d}} \quad (2)$$

Projecting the velocity vector onto the x and y directions yields,

$$v_x(t) = v \cdot \cos\theta(t) = v \cdot \cos\left(\sqrt{\frac{4\pi v \omega t}{d}}\right) \quad (3)$$

$$v_y(t) = v \cdot \sin\theta(t) = v \cdot \sin\left(\sqrt{\frac{4\pi v \omega t}{d}}\right) \quad (4)$$

The computational domain, illustrated in [Fig. 2](#), measures $20 \text{ mm} \times 20 \text{ mm} \times 6 \text{ mm}$ (length $\times$ width $\times$ depth) and is horizontally partitioned into an upper air domain and a lower T2 copper domain. A dual-scale mesh strategy is adopted to reconcile computational efficiency with local fidelity. The laser-irradiated and heat-affected region ($7 \text{ mm} \times 7 \text{ mm}$) is meshed with a uniform fine mesh of 0.04 mm , whereas the surrounding region ($13 \text{ mm} \times 13 \text{ mm}$) is meshed with a coarser grid of 0.2 mm . The spiral trajectory is computed for three complete revolutions. The air-domain top surface set as is imposed as a pressure outlet held at ambient temperature, and the copper surface is treated as a no-slip wall with coupled convective and radiative heat exchange. The laser-spot diameter matches the experimental settings. The thermophysical properties of T2 copper are listed in [Table 3](#).



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Fig. 2. The computational domain.

Table 3. Thermophysical properties of T2 copper.

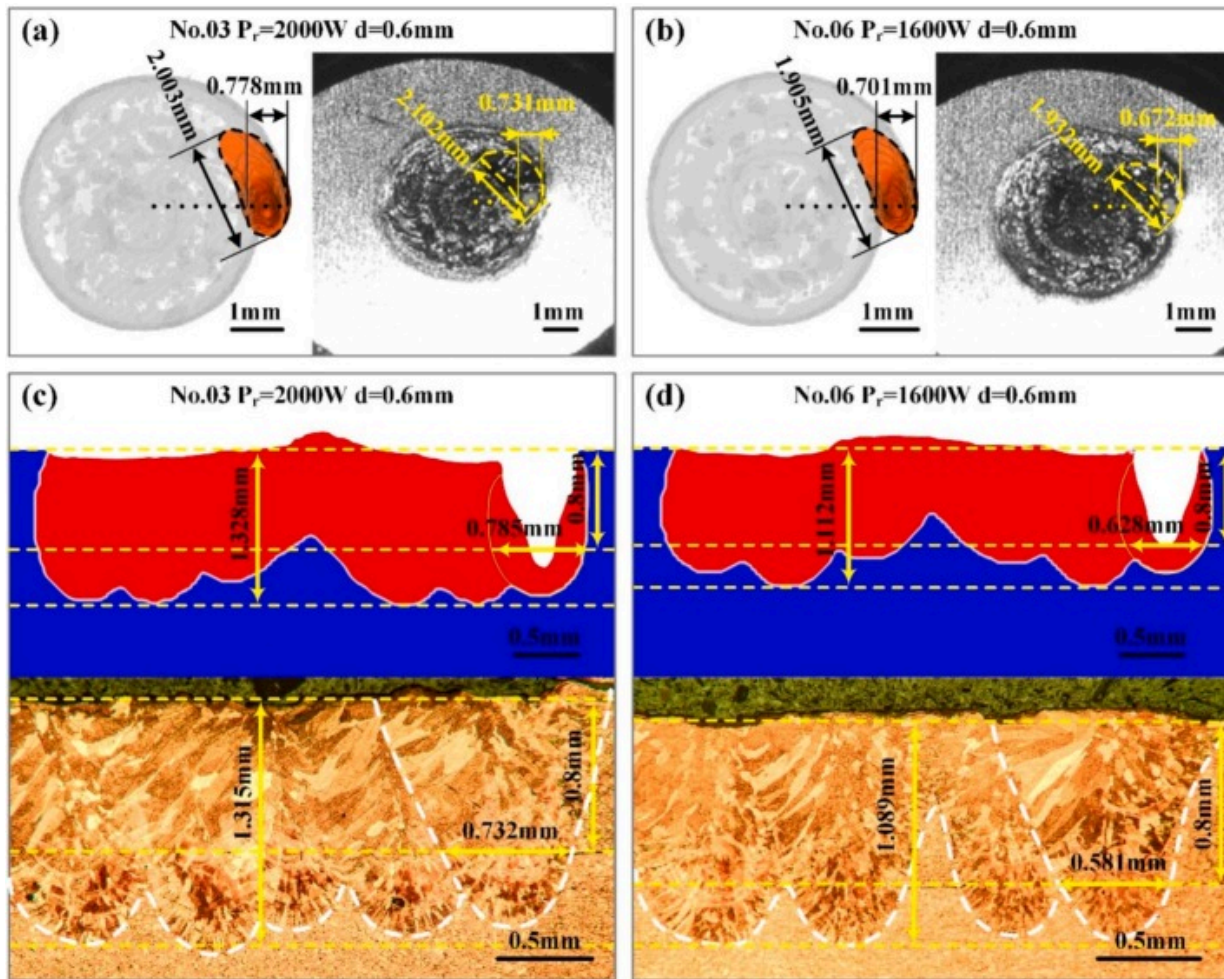
Parameter	Symbol	Unit	Value
Liquidus temperature	T_l	K	1356

Parameter	Symbol	Unit	Value
Solidus temperature	T_s	K	1338
Vaporization temperature	T_e	K	2835
Latent heat of fusion	H_f	J/kg	2.09e+05
Latent heat of vaporization	H_v	J/kg	4.73e+06
Solid density	ρ_s	kg/m ³	8920
Liquid density	ρ_l	kg/m ³	8130
Infrared laser absorption rate(@300 K)	α	–	2.1%
Coefficient of laser absorption variation with temperature	$d\alpha/dT$	–	0.0037%/K
Liquid viscosity (@1356 K)	η	kg/m*s	4.05e-03
Coefficient of liquid viscosity variation with temperature	$d\eta/dT$	kg/m*s*K	-2.68e-86
Surface tension (@1356 K)	γ	N/m	1.31
Coefficient of surface tension variation with temperature	$d\gamma/dT$	N/m*K	-2.3e-4
Specific heat of solid phase	c_s	J/kg*K	385
Specific heat of liquid phase	c_l	J/kg*K	494
Thermal conductivity of solid phase	λ_s	W/m*K	390
Thermal conductivity of liquid phase	λ_l	W/m*K	180

2.2.2. Model validation

To validate the model, the molten pool geometries and cross-sections obtained from case #03 and case #06 were compared against experimental measurements, as illustrated in Fig. 3. For case #03, the experimental results exhibited a molten pool length of 2.102 mm

and a width of 0.731 mm, compared to the simulated values of 2.003 mm and 0.778 mm, respectively. In terms of penetration depth and width at $Z = -0.8$ mm, the experiment reported 1.315 mm and 0.732 mm, versus 1.328 mm and 0.785 mm obtained from the simulation. Likewise, for case #06, the experimental molten pool geometry was measured, yielding a length of 1.932 mm and a width of 0.672 mm, whereas the simulation yielded 1.905 mm and 0.701 mm. At the $Z = -0.8$ mm interface, the penetration depth and width were recorded as 1.089 mm and 0.581 mm experimentally, in contrast to the simulated values of 1.112 mm and 0.628 mm. The favorable agreement observed across all key dimensions confirms that the numerical model is capable of accurately replicating the essential characteristics of the welding process.



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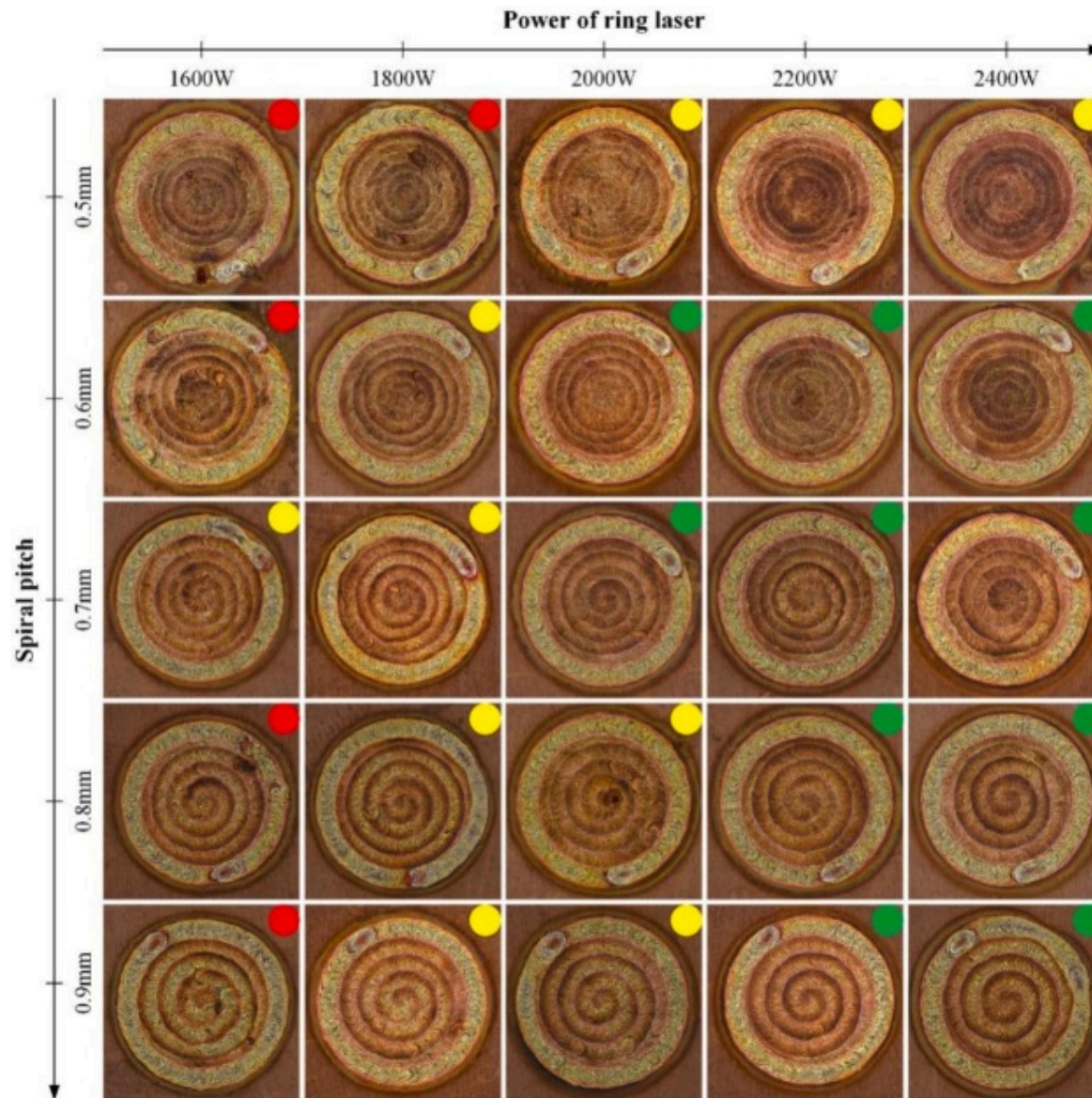
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Fig. 3. Comparison of numerical simulation and experimental results. (a) comparison of molten pool geometries for case #03, (b) comparison of molten pool geometries for case #06, (c) comparison of cross section for case #03, (d) comparison of cross section for case #06.

3. Result and discussion

3.1. Influence of process parameters on surface quality

Fig. 4 shows the surface appearance of welds obtained under 25 parameter combinations and ranks their quality through a three-level qualitative metric. This classification is based on both qualitative visual criteria and quantitative roughness thresholds that must be satisfied simultaneously. Specifically, “good” (marked in green) corresponds to an absence of craters and oxidation with regular fish-scale ripples, and requires $S_z \leq 350 \mu\text{m}$ and $S_q \leq 50 \mu\text{m}$; “medium” (marked in yellow) corresponds to small craters and light oxidation, with $S_z \leq 450 \mu\text{m}$ and $S_q \leq 55 \mu\text{m}$; and “poor” (marked in red) corresponds to coarse craters and heavy oxidation, with $S_z > 450 \mu\text{m}$ and $S_q > 55 \mu\text{m}$. With respect to power of ring, it exerts a monotonic influence. At 1600 W, the molten pool is inherently unstable, and spatter-induced craters and oxidation are ubiquitous. Incremental increasing the power systematically reduces these defects. Consequently, at 2200 W, all welds irrespective of spiral pitch are classified as good and spatter traces are virtually eliminated. With respect to spiral pitch, it exerts a non-monotonic influence. A 0.5 mm pitch intensifies local thermal accumulation and repeated remelting, aggravating surface flaws. Increasing the pitch to 0.7 mm provides optimal heat distribution and the best surface quality. A further increase to 0.8 mm lowers energy density, undermines molten pool stability, and causes defect resurgence, thereby degrading surface quality.



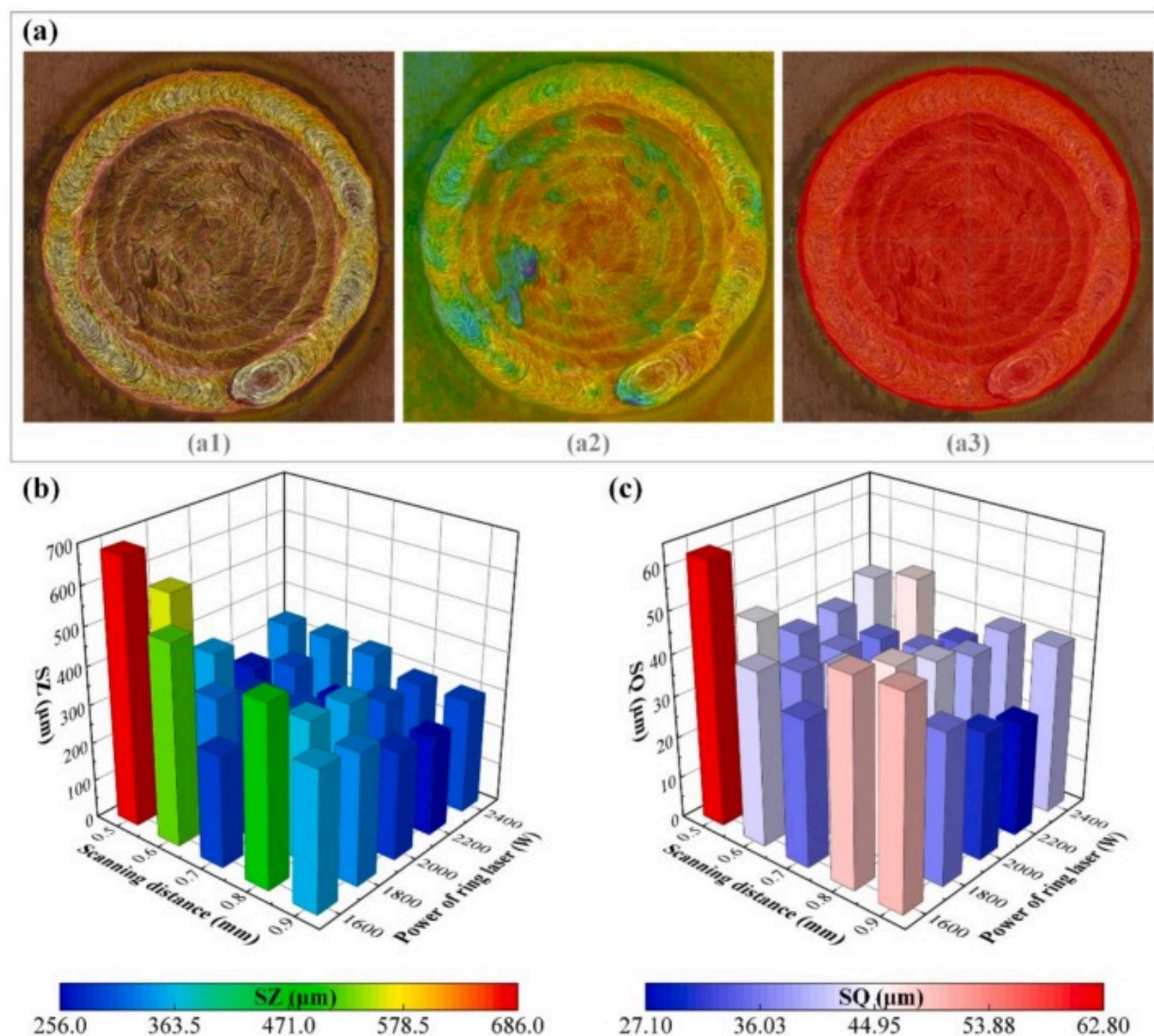
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Fig. 4. Weld surface morphology.

The three-dimensional topography of the weld was acquired using the Merge 3D module of a confocal microscope, enabling quantification of the influence of process parameters on surface roughness, with the measurement principle is illustrated in Fig. 5(a).

Specifically, height-field data within the weld region were extracted, and two roughness parameters were calculated, the maximum height SZ (indicating extreme irregularities) and the root-mean-square height SQ (which is insensitive to scratches, contamination, and measurement noise, yielding stable results). Fig. 5 (b) shows that SZ ranges from $257.25\text{ }\mu\text{m}$ ($P_r=2200\text{ W}$, $d=0.7\text{ mm}$) to $685.77\text{ }\mu\text{m}$ ($P_r=1600\text{ W}$, $d=0.5\text{ mm}$). The corresponding SQ values in Fig. 5 (c) span $27.19\text{ }\mu\text{m}$ ($P_r=2200\text{ W}$, $d=0.9\text{ mm}$) to $62.79\text{ }\mu\text{m}$ ($P_r=1600\text{ W}$, $d=0.5\text{ mm}$). With increasing power of ring, SZ and SQ decrease monotonically, at $d=0.6\text{ mm}$, elevating P_r from 1600 W to 2000 W reduces SZ by 30.29% and SQ by 18.62%. With increasing spiral pitch, SZ and SQ exhibit a non-monotonic “decrease-then-increase” trend, evidencing d as a non-linear regulator of spattering. Under $P_r=2000\text{ W}$, the optimal d interval is $0.6\text{--}0.7\text{ mm}$, deviations from this range cause a sharp roughness rise. For example, at $d=0.5\text{ mm}$ SZ and SQ rebound by 51.13% and 27.53%, respectively, while at $d=0.8\text{ mm}$ they rebound by 23.11% and 22.20%. These quantitative observations align excellently with the macroscopic surface quality ratings.



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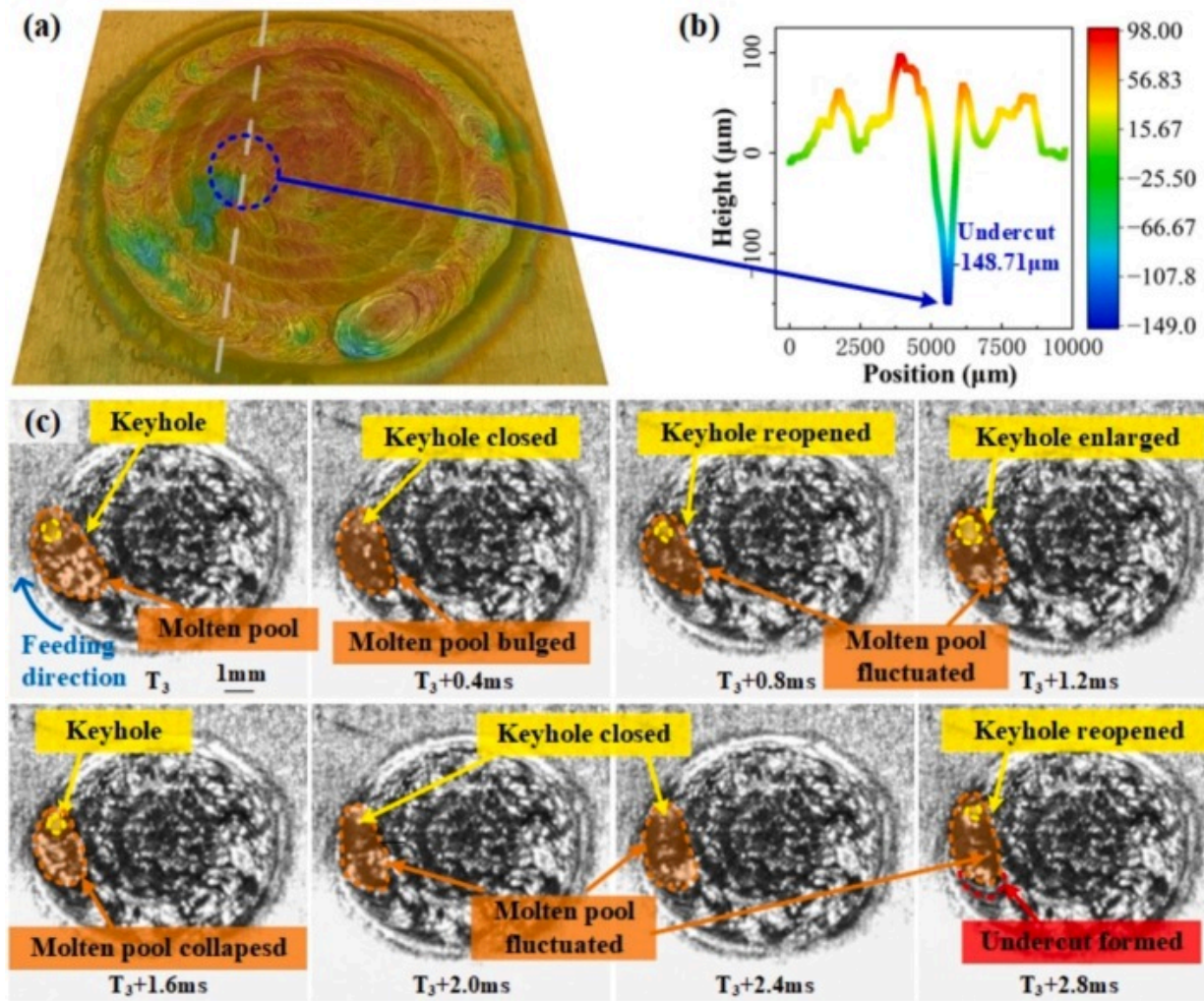
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Fig. 5. Surface roughness of the welds. (a) measurement principle of surface roughness, (a1) multi-focus surface imaging, (a2) generation of height map, (a3) measurement area, (b) maximum height SZ results, (c) root-mean-square height SQ results.

3.2. Process of spatter formation

3.2.1. Influence of power of ring on spatter formation

Fig. 6 characterizes the weld surface topography of specimen #06 ($P_r = 1600\text{ W}$, $d = 0.6\text{ mm}$) and the temporal evolution of a representative defect. Fig. 6(a) presents a three-dimensional reconstruction of the weld surface in which elevation is encoded by a blue–red color map, where blue indicates the lowest and red the highest elevation. A corresponding surface profile, extracted along the white dashed line in Fig. 6(a), is displayed in Fig. 6(b), which reveals that the maximum depth of the resultant pit is $254.29\text{ }\mu\text{m}$. The temporal evolution of this pit is captured in Fig. 6(c). At T_6 the keyhole is stably open and the molten pool surface is essentially quiescent. The keyhole aperture undergoes periodic radial oscillations that perturb the trailing region of the molten pool. By $T_6 + 1.6\text{ ms}$, the keyhole aperture becomes locally distorted. At $T_6 + 2.0\text{ ms}$, molten metal is expelled as a forward-projected spatter, whereupon the keyhole collapses. At $T_6 + 2.4\text{ ms}$, the central portion of the molten pool sags, and by $T_6 + 2.8\text{ ms}$, the keyhole re-opens while the collapsed material rapidly solidifies, leaving a surface depression.



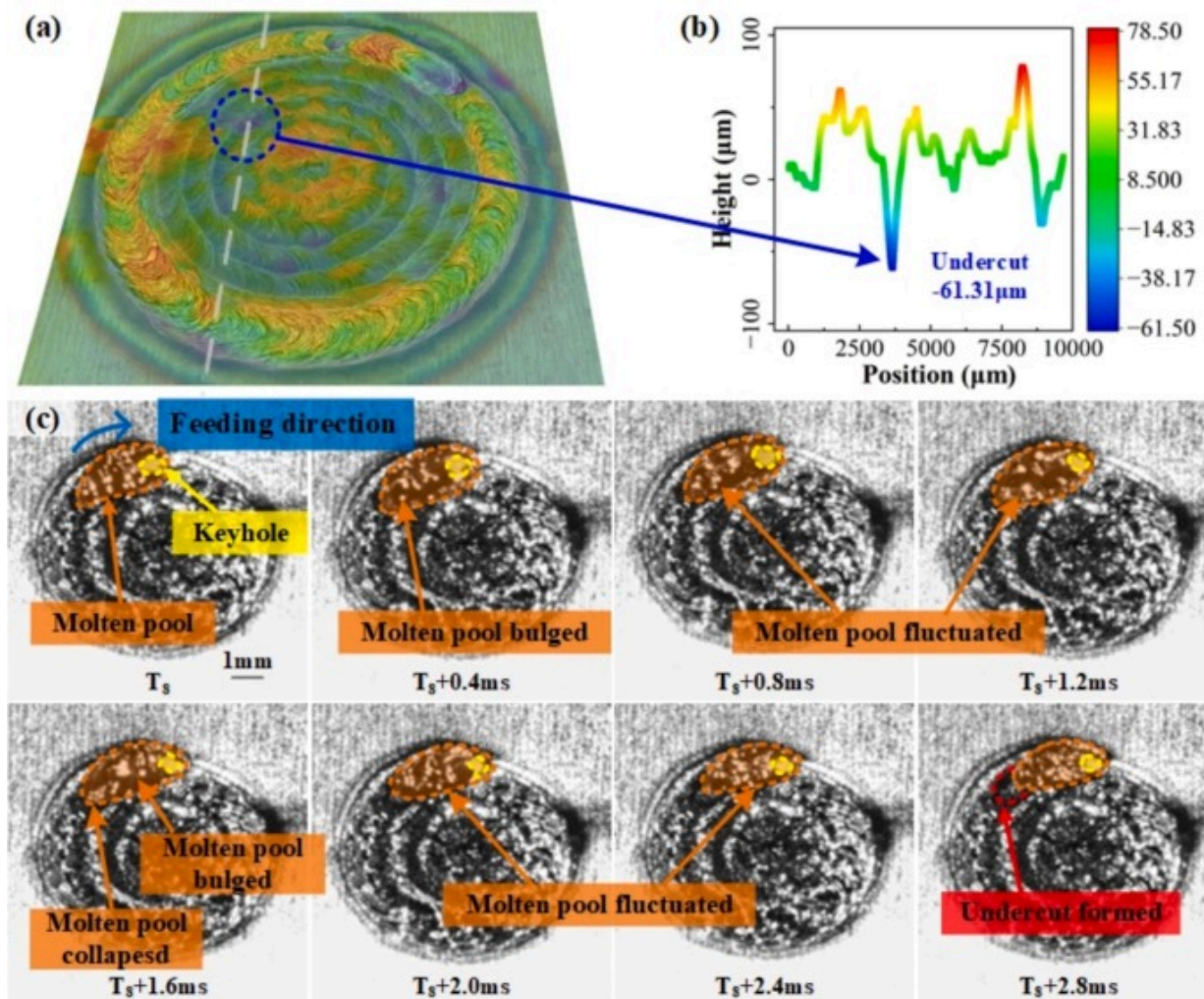
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Fig. 6. Surface topography and defect formation process of specimen #06. (a) surface topography of the specimen, (b) depth of the undercut, (c) process of undercut formation.

Fig. 7 characterizes the weld surface topography of specimen #08 ($P_r=2000W$, $d=0.6mm$) and the temporal evolution of a surface pit. Fig. 7(b) indicates a maximum pit

depth of $61.31\text{ }\mu\text{m}$. High-speed imaging (Fig. 7(c)) reveals that, beginning at T_8 , the keyhole undergoes periodic radial oscillations. At $T_8+0.4\text{ ms}$, a protuberance forms at the molten pool tail and propagates toward the keyhole. By $T_8+1.6\text{ ms}$, this region collapses inward. Solidification of the collapsed melt at $T_8+2.8\text{ ms}$ preserves the depression shown in Fig. 7(a). Throughout this sequence, the keyhole remains continuously open and no spatter is ejected.



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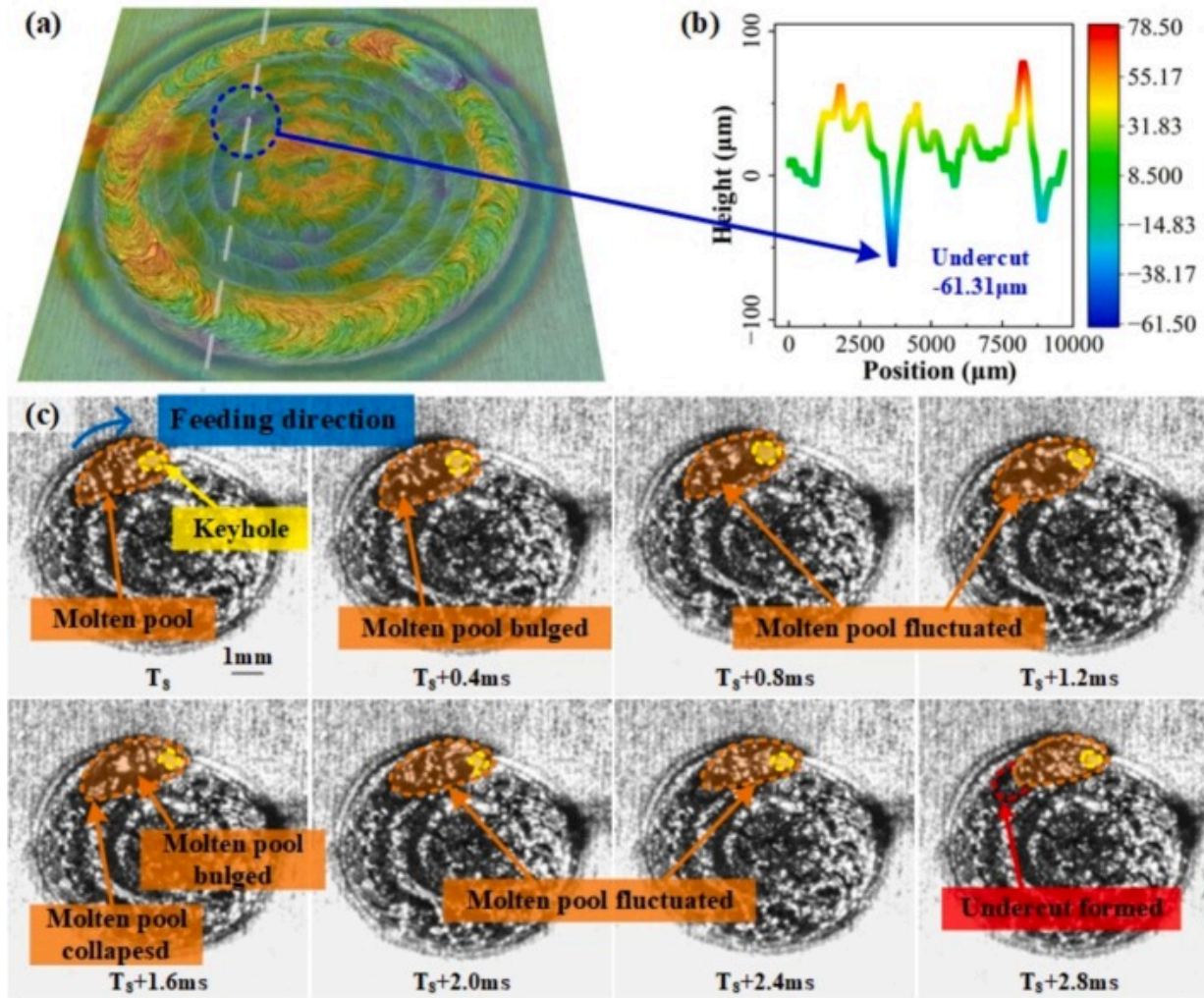
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Fig. 7. Surface topography and defect formation process of specimen #08. (a) surface topography of the specimen, (b) depth of the undercut, (c) process of undercut formation.

At a constant spiral pitch of 0.6 mm, increasing the power of ring from 1600 W to 2000 W markedly stabilizes the keyhole and molten pool, suppresses spatter probability and concurrently decreases weld-surface roughness.

3.2.2. Influence of spiral pitch on spatter formation

Fig. 8 characterizes the weld surface topography of specimen #03 ($P_r = 2000$ W, $d = 0.5$ mm) and the temporal evolution of a surface depression measuring $148.71 \mu\text{m}$ deep. At T_3 , the keyhole is stably opened, analogous to the behavior observed in specimen #08. At $T_3 + 0.4$ ms, a pronounced bulge emerges at the mid-rear region of the molten pool, accompanied by keyhole collapse. The keyhole re-opens at $T_3 + 0.8$ ms. Thereafter, up to $T_3 + 2.4$ ms, it undergoes periodic radial oscillations while the molten pool exhibits large-amplitude fluctuations that markedly exceed those recorded for specimen #08. Solidification of the perturbed melt as the beam advances forms the pit illustrated in Fig. 8(c). Compared with specimen #08, reducing the spiral pitch to 0.5 mm lengthens the total scan trajectory and increases the total heat input, intensifying keyhole and molten pool instabilities and increasing the resultant surface roughness. Nevertheless, no measurable spatter is generated.



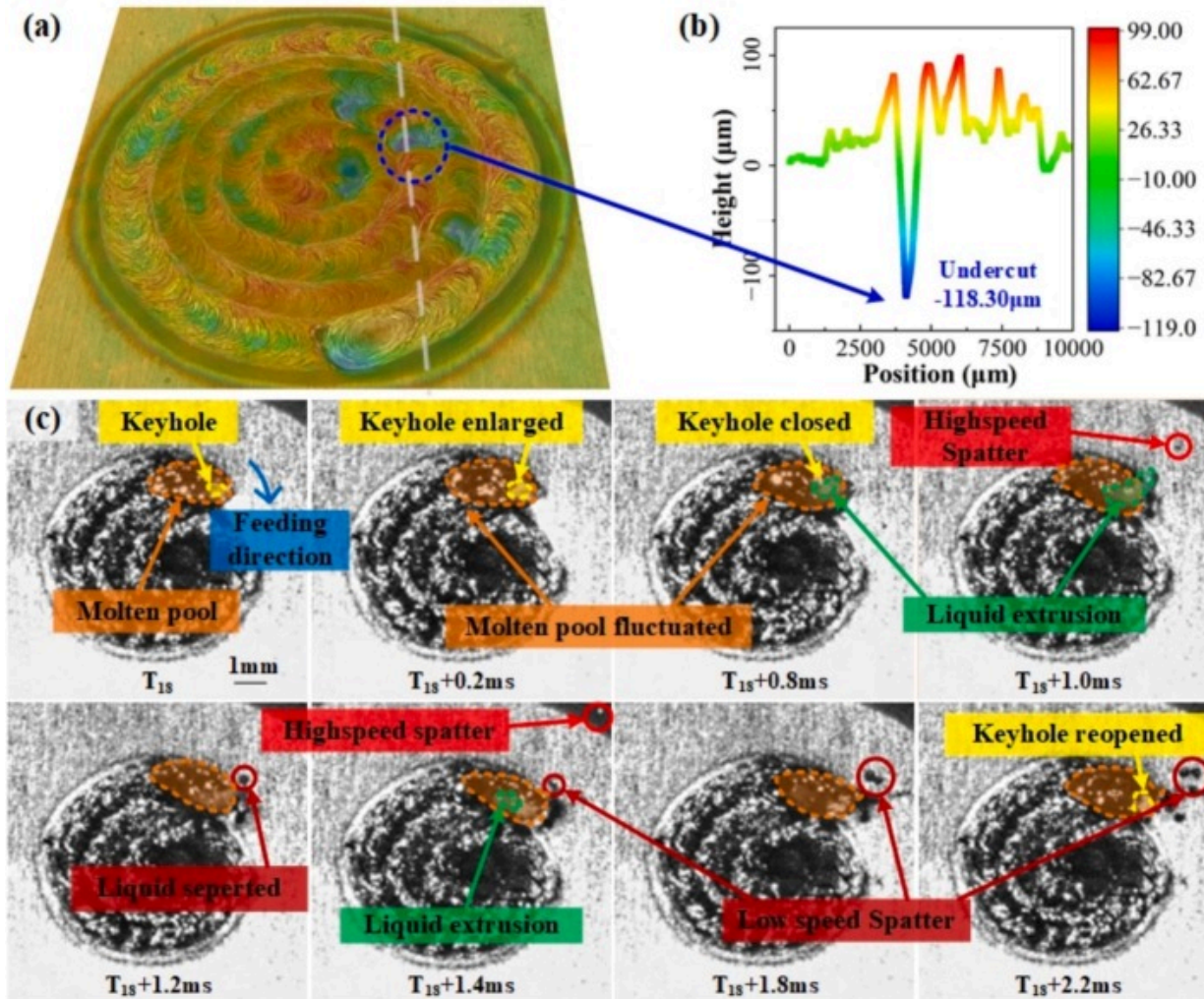
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Fig. 8. Surface topography and defect formation process of specimen #03. (a) surface topography of the specimen, (b) depth of the undercut, (c) process of undercut formation.

Fig. 9 characterizes the weld surface topography of specimen #18 ($P_r=2000W$, $d=0.8\text{ mm}$) and the temporal sequence that produces a surface pit measuring $118.30\mu\text{m}$ deep. At T_{18}

the keyhole is stably opened. Between $T_{18} + 0.2$ ms and $T_{18} + 1.0$ ms, the keyhole first expands then collapses, with a protuberance at the molten pool tail migrating toward the keyhole, forming an upward liquid jet, and rupturing to generate a high-velocity spatter. From $T_{18} + 1.2$ ms to $T_{18} + 1.8$ ms, a second jet rises from the molten pool center, detaches, and produces a larger but slower droplet. The keyhole re-opens at $T_{18} + 2.2$ ms. The mass loss induced by the consecutive spatters and the attendant oscillations of the molten pool result in the final depression.



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Fig. 9. Surface topography and defect formation process of specimen #18. (a) surface topography of the specimen, (b) depth of the undercut, (c) process of undercut formation.

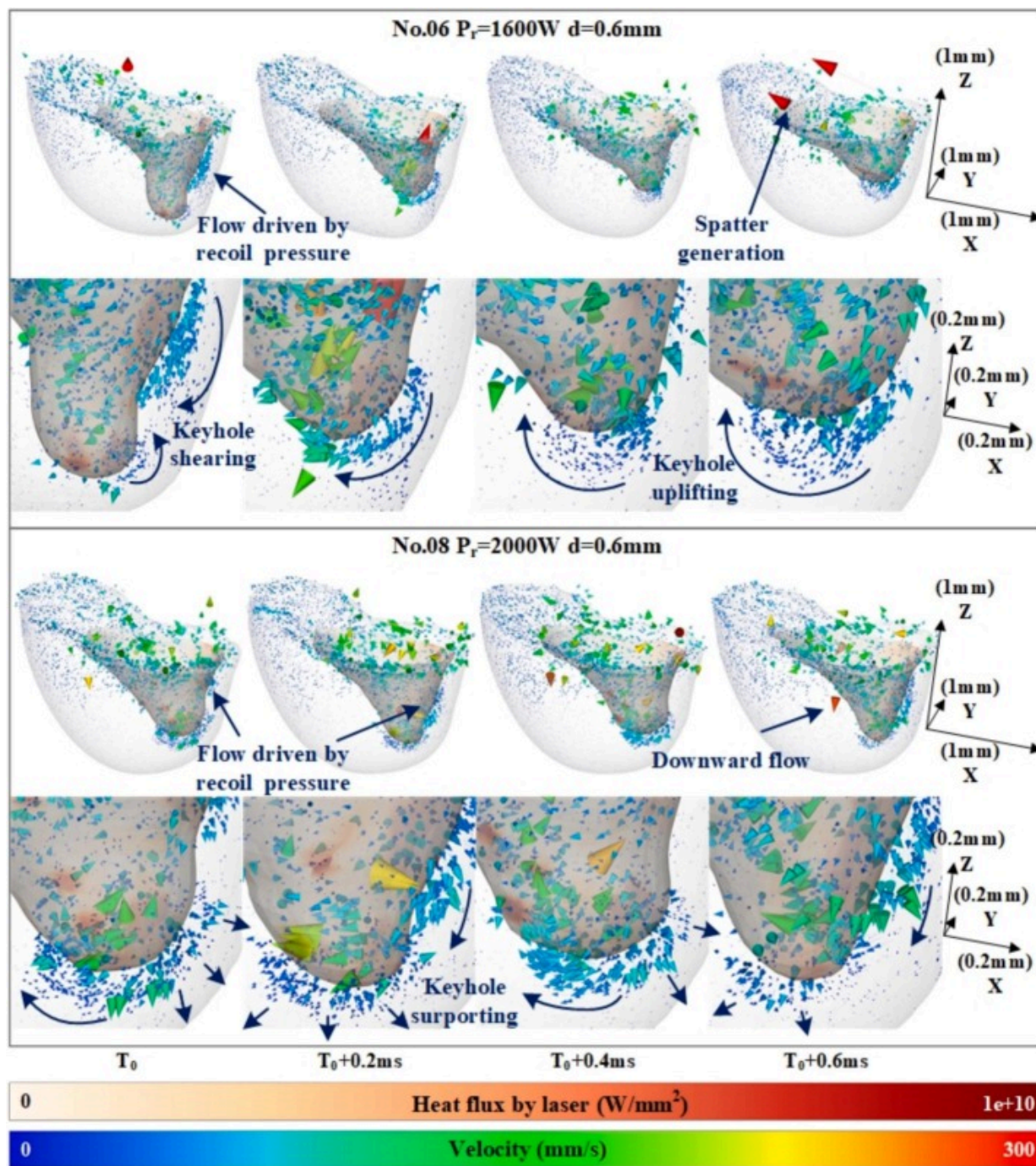
At a fixed power of ring of 2000 W, a spiral pitch of 0.5 mm yields a high areal heat input, enlarging the remelted zone and intensifying molten pool convection. Consequently, surface roughness increases, yet the keyhole remains sufficiently stable to suppress spatter. Increasing the pitch to 0.6 mm enhances pool stability and reduces roughness. A further increase to 0.8 mm markedly attenuates the cyclic preheat inherent to spiral scanning, inducing periodic keyhole collapse and amplified molten pool oscillations, which ultimately trigger spatter.

3.3. Mechanism of spatter formation

3.3.1. Mechanism of power of ring on spatter generation

[Fig. 10](#) compares the instantaneous keyhole dynamics and molten pool flow fields of case #06 ($P_r=1600$ W, $d=0.6$ mm) and case #08 ($P_r=2000$ W, $d=0.6$ mm) at an equivalent temporal reference. At 1600 W, the keyhole at $t=T_0$ exhibits a high depth-to-diameter ratio with pronounced mid-height constriction. This slender geometry results in multiple reflections that lead to enhanced energy deposition on the upper anterior keyhole wall. The localized recoil pressure generated there drives a downward flow along the anterior wall, forming a clockwise vortex. Simultaneously, at the keyhole bottom, the recoil pressure induced by intensified local evaporation expels the melt radially outward and upward, establishing a counter-clockwise vortex near the root region. By $T_0+0.2$ ms, these two counter-rotating vortices converge to generate a localized shear zone that constricts the keyhole and reduces its depth. This shear-localization mechanism, as corroborated by Lee's numerical research [35] and Schricker's synchrotron-radiography data [36], precipitates root contraction and a concomitant reduction in keyhole depth. The resultant energy concentration at the bottom surface intensifies local evaporation. This vapor shear

acts in concert with the basal clockwise vortex to produce a coupled uplift and constriction deformation of the keyhole root at $T_0 + 0.4$ ms. The subsequent beam convergence on the keyhole crown amplifies the local vapor shear until it surpasses the surface tension threshold, triggering spatter generation at $T_0 + 0.6$ ms. Increasing the ring power to 2000 W widens the keyhole entrance and lowers its aspect ratio, which reduces multiple reflections on the upper wall and allows the beam to propagate more directly to the root. The enhanced recoil pressure drives radial melt expulsion, creating a quasi-steady hydrodynamic “supporting effect” on the keyhole walls that suppresses upward melt motion and thereby markedly reduces spattering propensity.

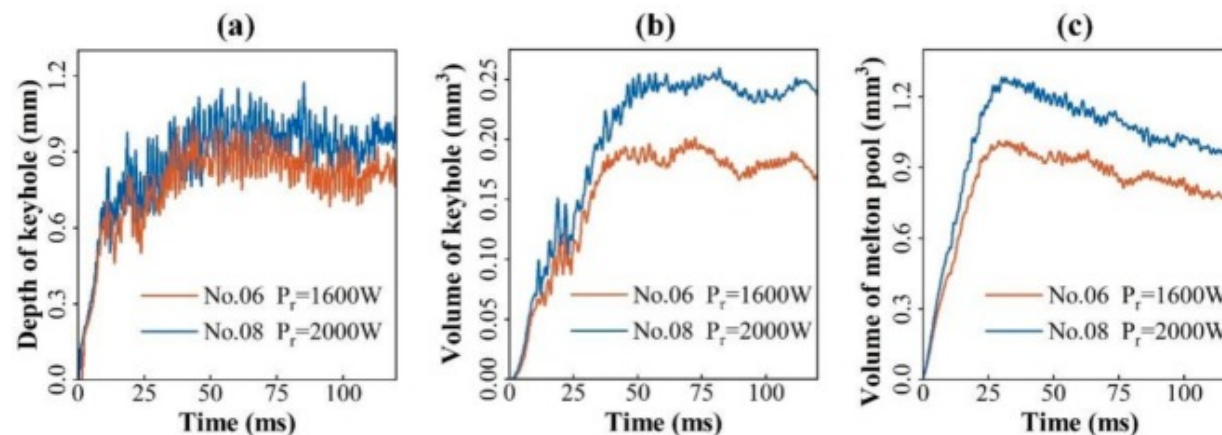


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Fig. 10. Dynamic behavior of the keyhole and molten pool flow at different powers of ring.

Fig. 11 quantifies the temporal evolution of keyhole depth, keyhole volume, and molten pool volume measured 120 ms after weld initiation. Elevating the power of ring concurrently enlarges the keyhole geometric scale (depth and volume), enabling the laser beam to reach the root of keyhole. This expansion homogenizes the energy flux incident on the keyhole wall and establishes an unobstructed channel for the rapid efflux of metallic vapor. Simultaneously, the enlarged molten pool volume increases the liquid inertial mass and viscous damping, thereby providing an inertial buffer that attenuates the periodic fluctuations in recoil pressure impinging on the keyhole wall. Collectively, these effects contribute to greater molten pool stability, thereby minimizing the occurrence of spatter.

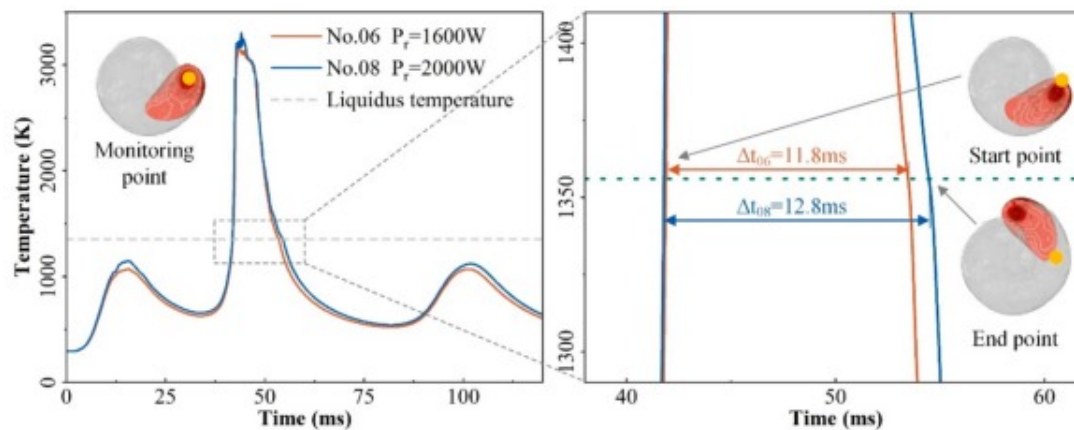


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Fig. 11. Temporal evolution of keyhole and molten pool metrics at varying powers of ring, (a) keyhole depth, (b) keyhole volume, (c) molten pool volume.

Furthermore, the absorptivity of T2 copper to infrared laser radiation increases exponentially with temperature. Fig. 12 presents the temporal evolution of surface temperature at the 4π position of the welding trajectory. Owing to the cyclic nature of the spiral path, the substrate undergoes preheating prior to beam arrival and slow cooling after its passage. Raising the power of ring from 1600W to 2000W elevates the mean temperature at the monitoring point and prolongs the time above the liquidus temperature from 11.8 ms to 12.8 ms, an increase of 8.47%, thereby markedly increasing laser absorptivity through enhanced surface temperature. The extended molten pool lifetime enhances the probability of internal vapor venting, which suppresses spatter driven by violent evaporation and provides additional time for the back-filling of liquid metal along the rear keyhole wall. Consequently, the recoil velocity of the liquid column generated by transient keyhole collapse is diminished, reducing the kinetic energy available for ejecting surface droplets.



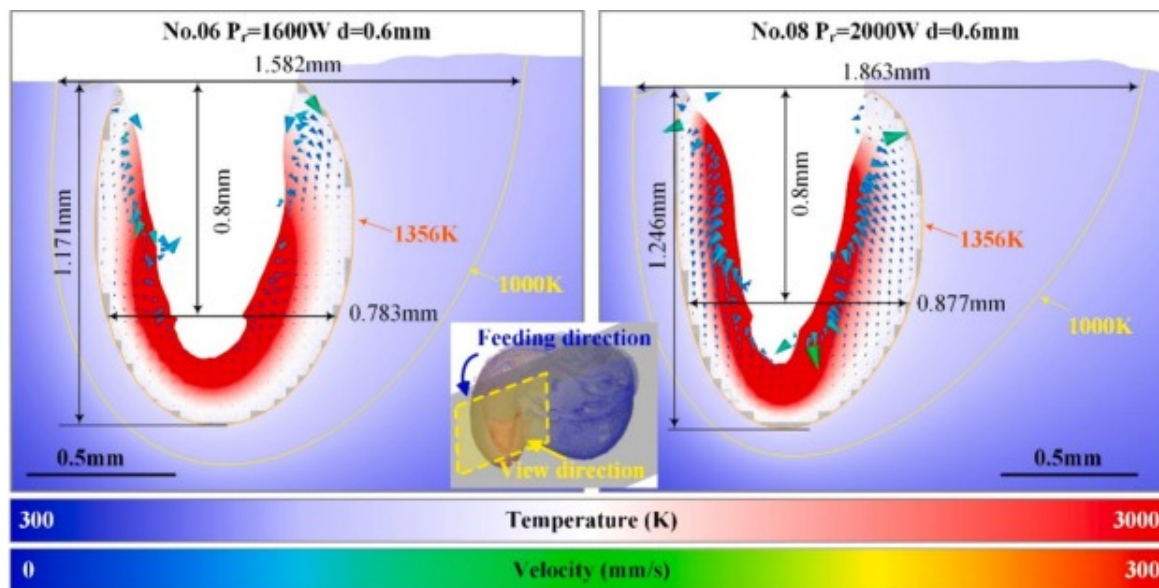
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Fig. 12. Temporal evolution of surface temperatures at varying powers of ring.

Fig. 13 illustrates the instantaneous radial cross-section of the keyhole and molten pool including its geometry, temperature field, and flow field, as the laser passes the monitoring point defined in Fig. 12. In agreement with numerical simulations by Yan [14],

the pool is radially asymmetric. The spiral center side retains a higher temperature because it coincides with the scan origin and is continuously heated by successive turns. As the power of ring increases from 1600 W to 2000 W, the molten pool depth grows from 1.171 mm to 1.246 mm (an increase of 11.55%), the surface 1000 K isotherm widens from 1.582 mm to 1.863 mm (an increase of 21.92%), and the effective fusion width at the $Z = -0.8$ mm interface rises from 0.783 mm to 0.877 mm (an increase of 12.01%). These variations indicate that the elevated P_r markedly reduces the temperature gradient across the molten pool and the adjacent base metal, thereby significantly weakening the thermocapillary-driven Marangoni convection. Simultaneously, the increased power of ring homogenizes the longitudinal keyhole temperature field. Recoil pressure expels molten metal laterally from the keyhole aperture, exerting a “flaring” action that separates the upper Marangoni current from the lower vortex, thereby suppressing necking-induced collapse of the keyhole crown.



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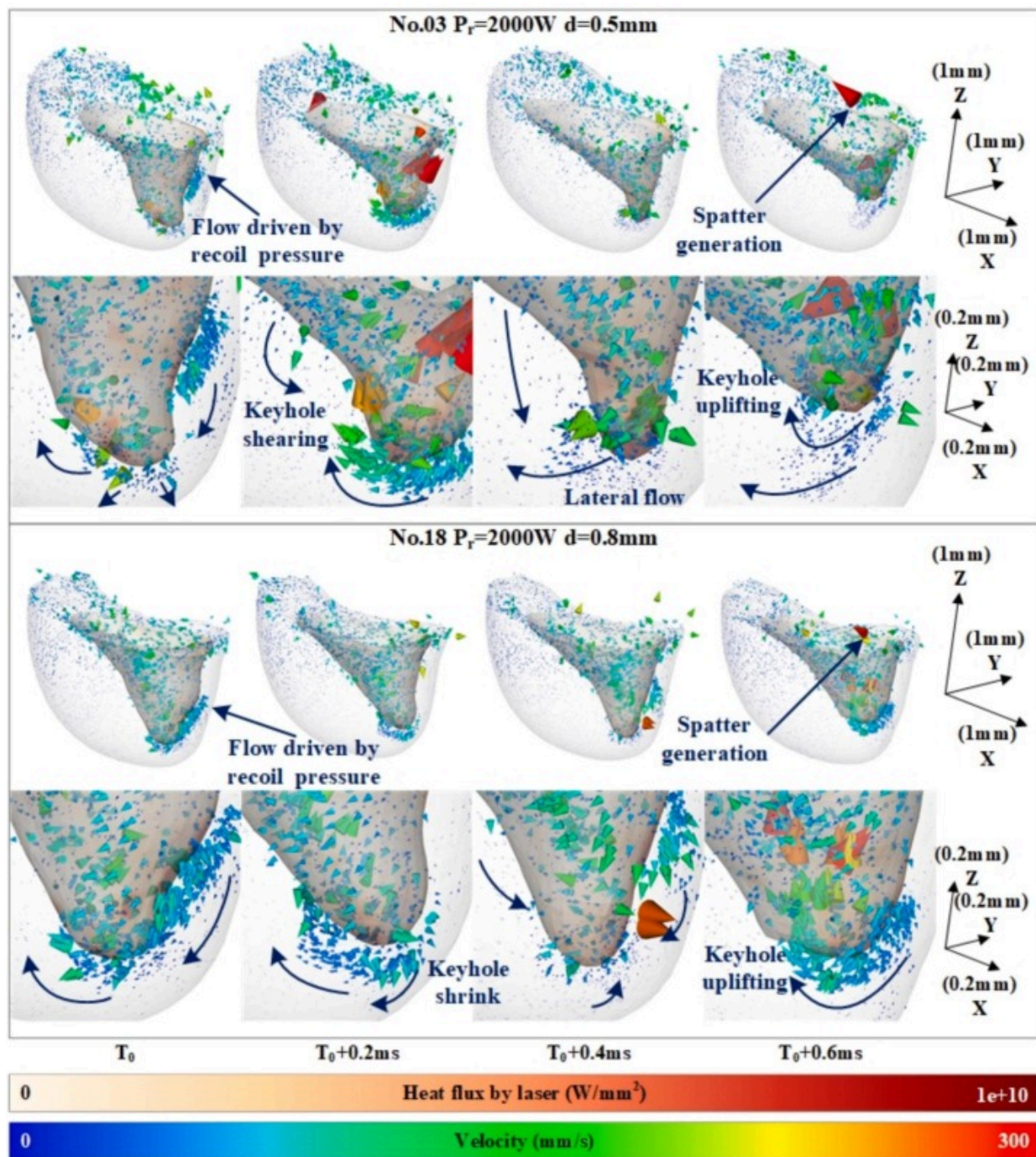
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Fig. 13. Temperature and flow fields of the radial cross-section at varying powers of ring.

3.3.2. Mechanism of spiral pitch on spatter generation

Fig. 14 compares the instantaneous keyhole dynamics and molten pool flow for case #03 ($P_r=2000\text{ W}$, $d=0.5\text{ mm}$) and case #18 ($P_r=2000\text{ W}$, $d=0.8\text{ mm}$) at the same angular position along the spiral path. For a given spiral outer diameter, a larger pitch reduces the total beam trajectory length and, consequently, the total energy input, thereby influencing both the preheating duration and local enthalpy accumulation. At a pitch of 0.5 mm , the closely overlapping passes generate intense preheating, which enhances absorptivity and produces a molten pool with pronounced curvature. Owing to this curved geometry, a fraction of the laser energy is deposited onto the keyhole's inner wall near the spiral center. The resultant localized evaporation induces a recoil-pressure-driven lateral flow perpendicular to the feeding direction of the laser beam. This lateral flow interacts with and merges into the upward flow emanating from the keyhole bottom, collectively forming a strong transverse vortex that exerts a concentrated lateral impact on the keyhole wall. By $T_0+0.2\text{ ms}$, intensified evaporation amplifies this recoil-pressure-driven lateral motion, leading to a pronounced transverse oscillation of the keyhole. At this stage, the keyhole exhibits upward necking while maintaining a wide aperture at the top, allowing the beam to reach the keyhole root. This provokes violent evaporation at the bottom. The sustained action of this transverse vortex promotes further keyhole constriction by $T_0+0.4\text{ ms}$, culminating in a sudden depth reduction at $T_0+0.6\text{ ms}$. Concurrently, the energy density at the keyhole crown rises abruptly, the basal flow becomes increasingly chaotic, pool fluctuations intensify, and the likelihood of spatter increases substantially. Increasing the pitch to 0.8 mm attenuates the preheating effect. Under this condition, the keyhole exhibits a high depth-to-diameter ratio. At the initial time T_0 , the laser beam simultaneously irradiates both the keyhole root and the leading wall. A fraction of the beam energy is deposited onto the mid-keyhole region; however, the resulting recoil pressure is insufficient to radially expel the molten material and thereby expand the keyhole aperture. A weak vortex initiates at the keyhole bottom by $T_0+0.2\text{ ms}$, and the keyhole starts to contract at $T_0+0.4\text{ ms}$. Compared to case #08 ($P_r=2000\text{ W}$, $d=0.6\text{ mm}$), the aperture at the keyhole root is narrower. By $T_0+0.6\text{ ms}$, the keyhole undergoes global upward shrinkage, resulting in an irregular root morphology

accompanied by substantial upward momentum, which ultimately triggers droplet ejection.

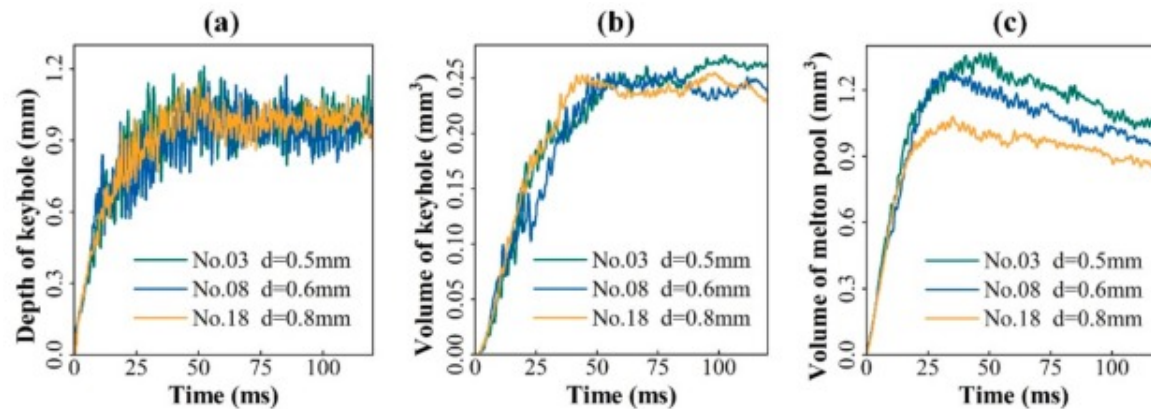


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Fig. 14. Dynamic keyhole behavior and molten pool flow at varying spiral pitches.

Fig. 15 quantifies the temporal evolution of keyhole depth, keyhole volume and molten pool volume 120 ms after weld initiation for three spiral pitches. Although keyhole depth and volume remain virtually constant across the pitch range, increasing the spiral pitch monotonically reduces the molten pool volume and markedly attenuates keyhole oscillations. This behavior is attributed to the diminished cyclic preheating effect inherent to wider spiral trajectories and the associated reduction in total energy input, which collectively enhance the stability of the molten pool surrounding the keyhole. However, the resultant decrease in melt volume reduces the inertial and viscous damping capacity of the pool, rendering it more susceptible to periodic fluctuations in recoil pressure and thereby weakening the “inertial buffering” effect. This mechanism also accounts for the non-monotonic influence of spiral pitch on spatter behavior.

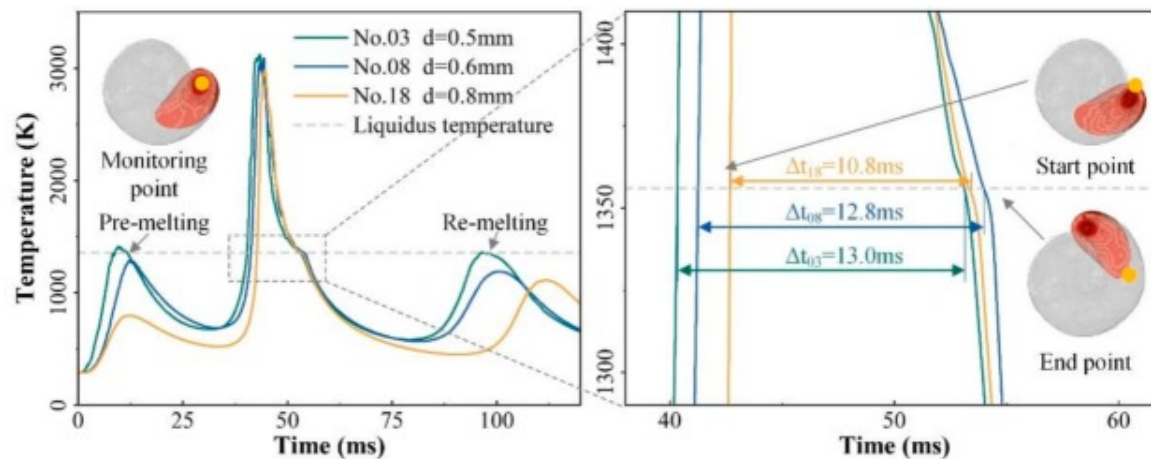


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Fig. 15. Temporal evolution of keyhole and molten pool metrics at varying spiral pitches, (a) keyhole depth, (b) keyhole volume, (c) molten pool volume.

Fig. 16 presents the temporal evolution of surface temperature at the 4π position of the welding trajectory. At a spiral pitch of 0.5 mm, the mean temperature is highest and the monitoring point remains above the melting point for 13.0 ms. Owing to the tight spiral, this location is repeatedly melted. In the absence of shielding gas, a surface oxide film inevitably forms. During subsequent passes, the laser repeatedly irradiates this oxide, producing transient absorptivity variations that destabilize the molten pool, an effect that accounts for the marked deterioration of surface quality in specimen #03 (Fig. 8). Increasing the pitch to 0.8 mm lowers the mean temperature and thus the initial absorptivity, shortening the melting interval to 10.8 ms, which represents a 15.63% decrease compared with the 0.6 mm pitch. The reduced liquid lifetime impedes complete melt spreading and backfilling, leaving the keyhole wall inadequately damped. Under cyclic recoil pressure, the keyhole undergoes vigorous pulsations, and the attendant vapor over-pressure ultimately ejects molten metal.



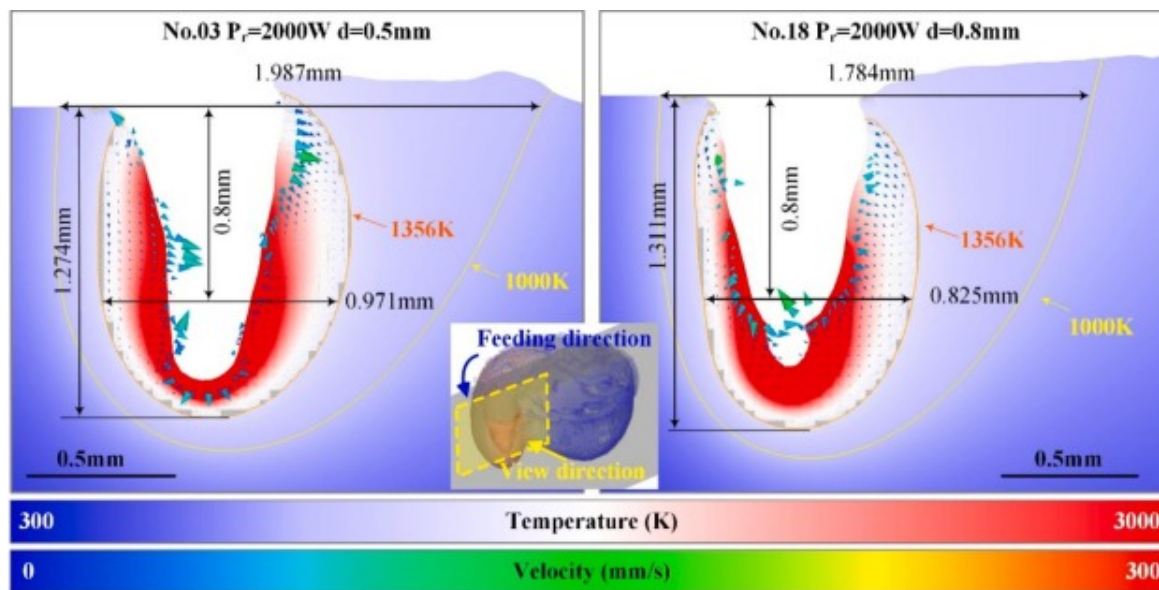
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Fig. 16. Temporal evolution of surface temperatures at varying spiral pitches.

Fig. 17 illustrates the instantaneous radial cross section of the keyhole-pool system, including its geometry, temperature field, and flow field, as the laser passes the

monitoring point defined in Fig. 16. Increasing the spiral pitch from 0.5 mm to 0.8 mm enlarges the molten pool depth from 1.274 mm to 1.311 mm (an increase of 2.90%), while narrowing the surface 1000 K isotherm from 1.987 mm to 1.784 mm (a decrease of 11.73%) and reducing the fusion width at the Z=-0.8 mm interface from 0.971 mm to 0.825 mm (an increase of 15.04%). At 0.5 mm pitch, the high spiral curvature drives radially inward flow, causing the lateral keyhole oscillation shown in Fig. 10. Compared with the 0.6 mm case in Fig. 13, the 0.8 mm pitch yields a 5.22% deeper pool and a 5.93% narrower effective fusion width, thereby increasing the depth-to-width ratio and the temperature gradient across the pool and adjacent base metal. The intensified Marangoni convection compresses the high-temperature zone to the pool bottom. Localized overheating sharply raises the evaporation rate and recoil-pressure peak. Once this pressure exceeds the sum of hydrostatic head and surface tension, violent ejection is triggered and the spatter propensity increases.



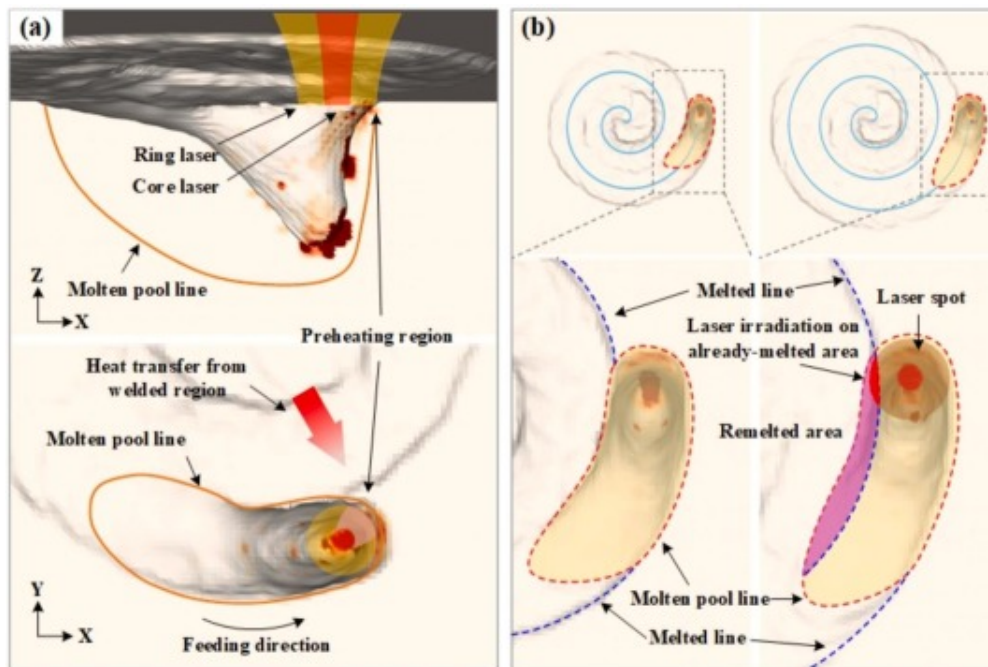
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Fig. 17. Temperature and flow fields of the radial cross-section at varying spiral pitches.

3.4. Spatter suppression mechanism and experimental verification

Experimental and numerical results consistently demonstrate that synergistic modulation of P_r and d markedly suppresses spatter during RMLSSW of T2 copper. Elevating P_r while reducing d intensifies substrate preheating, as shown in Fig. 18(a), thereby enhancing laser absorptivity and energy coupling efficiency [9]. It additionally lowers the molten pool temperature gradient, attenuates Marangoni convection driven by thermocapillary forces, and stabilizes molten pool flow [20]. Concurrently, the keyhole geometry is optimized, local vapor shear and vortex coupling are suppressed, leading to effective spatter mitigation [24]. Nevertheless, at excessively small d , laser irradiation impinges on the already melted metal surface as shown in Fig. 18(b). This readily induces remelting [28] and causes the molten pool flow to transition from steady laminar to turbulent, which re-initiates spatter formation [37]. Given that increasing P_r also widens the weld [24], [31], a processing window of $P_r=2000\text{--}2400\text{ W}$ and $d=0.55\text{--}0.70\text{ mm}$ is proposed. This ensures adequate preheating while preventing excessive heat input and remelting.



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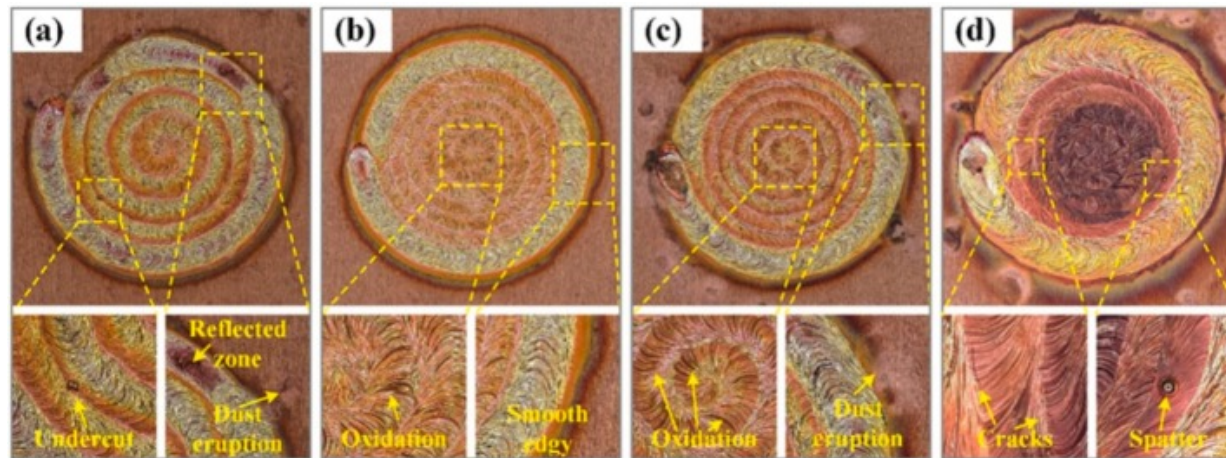
Fig. 18. Schematic diagram of suppression methods, (a) Schematic diagram of preheating mechanism, (b) Schematic diagram of remelting mechanism.

To validate the foregoing proposition, validation experiments were designed and executed (Table 4), with the resultant macro-morphologies displayed in Fig. 19. Specimen #a, characterized by a relatively low overall heat input, exhibited an oxide-free surface. However, the preheating supplied by the P_r and d proved insufficient, leaving black ash ejection traces around the weld seam, spatter-induced craters on the surface, and lack-of-penetration zones attributable to high reflectivity. Specimen #b, benefiting from an intensified cumulative preheating regime, presented a smooth, oxide-free surface devoid of ejection traces and was identified as the optimum condition. In specimen #c, the elevated P_r led to markedly increased oxidation and caused the reappearance of minor black ash ejection traces adjacent to the weld. Specimen #d, subjected to excessive heat input and surface temperatures surpassing the boiling point, experienced violent spatter, resulting in macroscopic craters accompanied by dark oxide bands and micro cracks. These findings substantiate the rule that moderate preheating suppresses spatter whereas overheating provokes it, and establish that the upper limit of P_r is dictated by the surface boiling temperature [15], thereby providing transferable process criteria and optimization boundaries for scanning welding of highly reflective materials.

Table 4. Experimental validation parameters.

No.	Power of center (P_c , W)	Power of ring (P_r , W)	Spiral pitch (d , mm)	Welding velocity (v , mm/s)
a	3000	1900	0.75	200
b	3000	1900	0.55	200
c	3000	3000	0.55	200

No.	Power of center (P_c , W)	Power of ring (P_r , W)	Spiral pitch (d , mm)	Welding velocity (v , mm/s)
d	3000	4000	0.55	200



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Fig. 19. Experimental validation results.

4. Conclusion

A quantitative investigation of spattering behavior during RMLSSW of T2 copper is presented. A three-dimensional CFD model is developed to systematically elucidate the coupled mechanism through which P_r and d govern spatter generation. The principal conclusions are.

1. Experimental results demonstrate that, P_r and d exert a synergistic suppression effect on spatter within the parameter space investigated in this study. Surface quality improves monotonically with increasing P_r , with

spatter traces becoming virtually undetectable at the highest power level. In contrast, d influences surface quality in a non-monotonic manner, with optimal spatter suppression achieved within the range of 0.6–0.7 mm.

2. Simulations reveal that increasing P_r elevates substrate temperature and laser absorptivity, stabilizes melt flow around the keyhole, and reduces the probability of spatter initiation. Meanwhile, the evaporative recoil pressure expands and stabilizes the keyhole, effectively suppressing spatter caused by stochastic keyhole collapse.
3. A reduction in d suppresses spatter by intensifying preheating, however, when d is decreased to 0.5 mm, remelting-induced lateral molten pool flow markedly increases the spattering propensity.
4. A moderate preheating process window is proposed, $P_r=2000\text{--}2400\text{ W}$ and $d=0.55\text{--}0.7\text{ mm}$ maximize laser-material coupling efficiency while minimizing spatter.

CRediT authorship contribution statement

Wendi Wu: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Ping Jiang:** Project administration, Funding acquisition, Resources, Writing – review & editing. **Shaoning Geng:** Supervision, Funding acquisition, Project administration, Writing – review & editing. **Chu Han:** Visualization, Methodology. **Leshi Shu:** Formal analysis, Data curation. **Yilin Wang:** Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] J. Feng, P. Zhang, H. Yan, *et al.*
Application of laser welding in electric vehicle battery manufacturing: a review
Coatings, 13 (8) (2023)
[Google Scholar ↗](#)
- [2] M.J. Brand, P.A. Schmidt, M.F. Zaeh, *et al.*
Welding techniques for battery cells and resulting electrical contact resistances
J. Storage Mater., 1 (2015), pp. 7-14
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [3] F. Kaufmann, O. Hentschel, S. Roth, *et al.*
Effects of different laser wavelengths on process performance and weld seam properties in copper processing for electric vehicle applications
Opt. Laser Technol., 191 (2025)
[Google Scholar ↗](#)
- [4] V. Dimatteo, A. Ascari, P. Faverzani, *et al.*

The effect of process parameters on the morphology, mechanical strength and electrical resistance of CW laser-welded pure copper hairpins

J. Manuf. Process., 62 (2021), pp. 450-457



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [5] J. Ning, L.-J. Zhang, X.-Q. Yin, *et al.*
Mechanism study on the effects of power modulation on energy coupling efficiency in infrared laser welding of highly-reflective materials

Mater. Des., 178 (2019)

[Google Scholar](#)

- [6] A. Sadeghian, N. Iqbal
A review on dissimilar laser welding of steel-copper, steel-aluminum, aluminum-copper, and steel-nickel for electric vehicle battery manufacturing

Opt. Laser Technol., 146 (2022)

[Google Scholar](#)

- [7] L. Wan, Z. Tang, H. Yang, *et al.*
Achieving high robust laser conduction welding and enhanced joint conductivity in pure copper foil stacks using a flat-top blue laser

Opt. Laser Technol., 174 (2024)

[Google Scholar](#)

- [8] F. Kaufmann, R. Möttingdörfer, S. Roth, *et al.*
Investigation of coupling efficiency in laser beam welding of copper materials using brilliant infrared and green laser radiation

J. Manuf. Process., 131 (2024), pp. 2037-2050



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [9] D. Wu, J. Sun, Z. Li, *et al.*

Molten pool behaviors and energy absorption in coaxial hybrid blue-IR lasers welding of a copper material

Int. J. Therm. Sci., 184 (2023)

[Google Scholar ↗](#)

- [10] H.-C. Chen, G. Bi, M.L.S. Nai, *et al.*
Enhanced welding efficiency in laser welding of highly reflective pure copper

J. Mater. Process. Technol., 216 (2015), pp. 287-293



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [11] K. Lee, H. Ki
Enhancing coupling efficiency in laser keyhole welding of copper using femtosecond laser surface modification

Opt. Laser Technol., 139 (2021)

[Google Scholar ↗](#)

- [12] S. Huang, H. Yu, M. Lou, *et al.*
Effect of Ti interlayers on microstructure and mechanical properties of laser spiral fusion welded steel/aluminum lap joints

J. Mater. Res. Technol., 35 (2025), pp. 1752-1763



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [13] S. Huang, H. Yu, M. Lou, *et al.*
Formation mechanism of intermetallic compound in liquid steel and liquid aluminum by laser spiral fusion welding

J. Manuf. Process., 138 (2025), pp. 214-224



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [14] G. Yan, M.M. Pour, T.J. Rinker, *et al.*

Laser keyhole welding of dissimilar metals with spiral contours: metal mixing, microstructure, and mechanical strength

J. Manuf. Process., 141 (2025), pp. 1020-1030



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [15] J. Li, Y. Wang, P. Jiang, *et al.*
Spatter elimination mechanism and criterion for adjustable ring-mode laser welding of aluminum alloy

Int. J. Heat Mass Transf., 242 (2025)

[Google Scholar](#)

- [16] L. Mei, L. Lin, D. Yan, *et al.*
Metal spattering in laser scanning welding of T2 copper and welding quality

Opt. Lasers Eng., 161 (2023)

[Google Scholar](#)

- [17] Z. Wan, H.-P. Wang, J. Li, *et al.*
Effect of beam oscillation frequency on spattering in remote laser stitch welding of thin-gage zinc-coated steel with keyhole penetration

J. Mater. Process. Technol., 302 (2022)

[Google Scholar](#)

- [18] Z. Wang, L. Shi, W. Liao, *et al.*
Numerical and experimental study on spatter in oscillating laser-arc hybrid welding of aluminum alloy

J. Mater. Process. Technol., 332 (2024)

[Google Scholar](#)

- [19] W. Li, H. Dong, B. Zhang, *et al.*

The influence of adjustable ring-mode laser on the formation of intermetallic compounds and mechanical properties of ultra-thin Al-Cu lap welded joints

J. Mater. Process. Technol., 332 (2024)

[Google Scholar ↗](#)

- [20] Y. Zhang, J. Chen, W. Zhang, *et al.*
Study of spatter net forming mechanism and penetration mode under flexible ring mode laser welding
J. Mater. Res. Technol., 24 (2023), pp. 2213-2225
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [21] A.A. Lassila, E. Lundell, T. Andersson, *et al.*
Experimental and numerical investigation of process-induced recoil force in keyhole laser welding: Insights for validating multi-physics process simulations and modelling assumptions
J. Mater. Process. Technol., 341 (2025)
[Google Scholar ↗](#)
- [22] L. Wang, X. Gao, F. Kong
Keyhole dynamic status and spatter behavior during welding of stainless steel with adjustable-ring mode laser beam
J. Manuf. Process., 74 (2022), pp. 201-219
 [View PDF](#) [View article](#) [Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [23] M.R. Maina, Y. Okamoto, A. Okada, *et al.*
High surface quality welding of aluminum using adjustable ring-mode fiber laser
J. Mater. Process. Technol., 258 (2018), pp. 180-188
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [24] Z. Wang, M. Jiang, X. Chen, *et al.*
Mitigating spatters in keyhole-mode laser welding by superimposing additional ring-shaped beam
Opt. Laser Technol., 168 (2024)
[Google Scholar ↗](#)
- [25] M. Mohammadpour, L. Wang, F. Kong, *et al.*
Adjustable ring mode and single beam fiber lasers: a performance comparison
Manuf. Lett., 25 (2020), pp. 50-55
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [26] Q. Li, M. Luo, Z. Mu, *et al.*
Improving laser welding via decreasing central beam density with a hollow beam
J. Manuf. Process., 73 (2022), pp. 939-947
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [27] P. Yang, L. Jiang, T. Yu, *et al.*
Synergistic enhancement of weld stability and mechanical performance in laser welding of titanium alloy by coupling ring-shaped mode and beam oscillation
J. Mater. Process. Technol., 345 (2025)
[Google Scholar ↗](#)
- [28] Q. Li, B. Zhu, H. Li, *et al.*
Effect of spiral scan distance on the nanosecond-pulsed-laser lap joint of Al/Cu
Opt. Laser Technol., 158 (2023)
[Google Scholar ↗](#)
- [29] J.-H. Shin, H.-C. Noh, G.-D. Park

Effect of spiral welding path and laser power on weld in laser welding of aluminum tab for lithium-ion battery

Int. J. Adv. Manuf. Technol., 126 (3–4) (2023), pp. 1317-1327

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [30] J. Li, S. Geng, Y. Wang, *et al.*
Mitigation of porosity in adjustable-ring-mode laser welding of medium-thick aluminum alloy

Int. J. Heat Mass Transf., 227 (2024)

[Google Scholar ↗](#)

- [31] J. Li, P. Jiang, S. Geng, *et al.*
Numerical and experimental study on keyhole dynamics and pore formation mechanisms during adjustable-ring-mode laser welding of medium-thick aluminum alloy

Int. J. Heat Mass Transf., 214 (2023)

[Google Scholar ↗](#)

- [32] L. Yang, S. Geng, P. Jiang, *et al.*
Investigation on the keyhole/molten pool dynamic behavior during adjustable ring-mode laser welding of medium-thick aluminum alloy

Int. J. Therm. Sci., 196 (2024)

[Google Scholar ↗](#)

- [33] S. Geng, W. Yang, P. Jiang, *et al.*
Numerical study of keyhole dynamics and porosity formation during high-power oscillating laser welding of medium-thick aluminum alloy plates

Int. J. Heat Mass Transf., 194 (2022)

[Google Scholar ↗](#)

- [34] B. Xu, P. Jiang, Y. Wang, *et al.*
Multi-physics simulation of wobbling laser melting injection of aluminum alloy with SiC particles: SiC particles gradient distribution in fusion zone
Int. J. Heat Mass Transf., 182 (2022)

[Google Scholar ↗](#)

- [35] J. Lee, J. Jang, J. Park, *et al.*
Numerical and experimental analysis on the effect of ring beam-driven flow in coaxial dual-beam laser welding of aluminum alloy
Int. Commun. Heat Mass Transfer, 157 (2024)

[Google Scholar ↗](#)

- [36] K. Schricker, L. Schmidt, H. Friedmann, *et al.*
Characterization of keyhole dynamics in laser welding of copper by means of high-speed synchrotron X-ray imaging
Procedia CIRP, 111 (2022), pp. 501-506



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [37] Y. Wei, G. Chen, Z. Xiao, *et al.*
Microstructures and mechanical properties of pure copper manufactured by high-strength laser powder bed fusion
Opt. Laser Technol., 182 (2025)

[Google Scholar ↗](#)

[View Abstract](#)

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